Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Agent engineer who **built AgentProbe end-to-end** - production LangGraph platform with a 3-agent system (Orchestrator, RAG Tracer, Auto Fix), multi-step reasoning, dynamic tool calls, sandboxed local execution, and a 5-metric evaluation engine. Designed multi-step agent workflows coordinating across tools, APIs, and retrieval - the same pattern Vercel's Agent team ships for production incident investigation and autonomous remediation. 8+ years production engineering (Python, **TypeScript/JavaScript**, PostgreSQL) at Entrata - 120+ Agile sprints, 100K+ users, 1M+ records. Full-stack capability: FastAPI + React + AWS + Docker + GitHub Actions CI/CD. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

Agentic Systems Design (LangGraph) | Multi-Step Agent Workflows | LLM APIs (Anthropic, OpenAI, Groq)

Prompt Engineering & Model Evaluation | RAG + Vector Search (ChromaDB) | Tool Call Architecture | TypeScript / JavaScript / Python

Full-Stack (FastAPI + React) | CI/CD & Observability (GitHub Actions) | Production at Scale (1M+ records) | Open-Source Engineering

Sandboxed Execution (Docker)

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Built and delivered **20+ full-stack modules** (**Python, JavaScript/TypeScript, PostgreSQL**) across **120+ Agile sprints** for **100K+ monthly active users** - shipped independently and influenced architecture across Product, Engineering, and QA.
- Led **Docker-based containerization** for 30%+ of microservices - sandboxed execution environments for isolating services; cut microservices execution time by **50%**; maintained **99.5% uptime** for 6+ months via TDD/BDD.
- Improved production query performance by **30%** and reduced deployment times by **40%**; built **5+ internal debugging and automation tools** adopted across squads - reducing resolution time by **40%**.
- Applied **Python, PyTorch, and OpenAI API** to automate HUD income limit data transformation across **1.6M financial records** - production AI on regulated data at scale.
- Migrated client data to AWS S3 with Libsodium-encrypted pipelines - **100% PHI compliance, zero security incidents** across 100K+ users.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Mentored 30+ students in Python and databases; translated complex systems concepts across skill levels - same discipline as surfacing agent behavior in observability dashboards.

PROJECTS

AgentProbe: Production Multi-Agent Platform with Evaluation Engine OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

End-to-end agent platform built to investigate, evaluate, and fix AI system failures - the same mission as Vercel's Agent team for production infrastructure. **Agentic system design**: 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with defined action spaces, multi-step reasoning chains, and inter-agent state handoffs. **Tool call architecture**: dynamic routing across Anthropic, OpenAI, Groq, Ollama based on task type, latency, and capability - enables runtime capability acquisition per task. **Sandboxed local execution**: runs 100% locally via Docker and Ollama - zero cloud dependency, reproducible environment. **RAG pipeline**: ChromaDB + embeddings + retrieval quality scoring across multi-step runs. **Prompt engineering + model eval**: 5-metric LLM-as-judge engine (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD regression gates. **Observability**: 15-endpoint FastAPI backend + React dashboard surfacing per-run agent decisions. Open-source - used by other engineers. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: Domain-Specific NLP Query System for Medical Records

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

NLP pipeline mapping natural language queries to FHIR APIs via LLMs, BERT, and embeddings - HIPAA-compliant RBAC. **~60% retrieval time reduction.** Full-stack AI with tool integrations and domain-specific agent knowledge - the pattern Vercel's Agent uses to understand platform capabilities. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → evaluation (MDI + Permutation Importance) → stakeholder dashboard. **85%+ accuracy.** Systematic evaluation and deployment methodology.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

Agent & LLM: LangChain, LangGraph, Multi-Step Agent Workflows, Tool Call Architecture, LLM APIs (Anthropic, OpenAI, Groq, Ollama), Prompt Engineering, RAG Pipelines, ChromaDB, LLM-as-Judge Evaluation, Model Selection

Full-Stack & Infra: FastAPI, React, PostgreSQL, MySQL, Docker (sandboxed execution), AWS (S3, EC2, RDS, Glue), Azure, CI/CD (GitHub Actions), RabbitMQ, Microservices

ML & Data: PyTorch, scikit-learn, BERT, spaCy, Hugging Face, NumPy, Pandas, Feature Engineering, Statistical Modeling

Process: Open-Source Development, TDD/BDD, Integration Testing, Git, JIRA, Agile/Scrum